EVENT CHECK-IN PLATFORM

White-Label, Multi-Tenant SaaS

Product Requirements & Technical Design Document

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CONFIDENTIAL — Prepared for internal planning, development partners, and prospective investors

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1. Executive Summary

This document defines the product requirements and technical design for a white-label, multi-tenant event check-in platform. The platform is built to be licensed or sold to clients running large-scale events (10,000–15,000+ participants), enabling fast, secure, offline-capable participant check-in through single-use QR codes, real-time attendance visibility, and post-event reporting.

The platform is designed from the outset as a reusable SaaS product rather than a single-event build, with each client operating under their own branded environment and custom domain.

2. Product Overview

A web and mobile platform that manages event registration and check-in through unique, single-use QR codes. Participants register (self-service or bulk-uploaded by an admin) and receive a personal QR code by email. Gate staff use an in-app scanner — not the device's native camera — to validate and check in attendees at multiple entry points simultaneously, including in low or no connectivity conditions, with automatic reconciliation once devices reconnect.

3. Goals & Objectives

- Deliver a reusable, licensable product — not a single-event build
- Support 10,000–15,000+ participants per event without performance issues
- Enable simultaneous multi-gate check-in, fully functional offline
- Give each client their own branded environment with a custom domain
- Separate gate staff (scan-only) from full admin access for security and simplicity
- Provide email-first messaging, with SMS/WhatsApp as a paid add-on channel
- Deliver clean, exportable post-event reporting (CSV/PDF)

4. User Roles & Access

Role	Access & Permissions
Super Admin (Platform Owner)	Manages all client accounts, billing, custom domains, and platform-wide settings
Client Admin	Full access to their own tenant: create events, manage registrations, bulk upload, messaging, reports, and manage gate staff/admin users
Gate Staff	Scan-only login, scoped to one event and optionally one named gate; no access to registrant data beyond confirming a check-in
Participant	Registers via public link or is bulk-uploaded; receives QR code and messages by email (SMS/WhatsApp optional)

5. Core Features

5.1 Multi-Tenant, White-Label Architecture

- Each client gets an isolated tenant with their own data
- Custom domain/subdomain per client (e.g., checkin.clientname.com), with branding (logo, colors, sender identity)
- Super Admin console to provision new client accounts and domains

5.2 Registration

- Public self-registration link, customizable form fields per event
- Bulk upload via Excel/CSV, with validation, deduplication, and error reporting for bad rows
- Registration record includes Registered status, timestamp, and source (self / bulk)

5.3 QR Code Generation & Delivery

- One cryptographically signed, non-sequential QR code per participant
- Single-use — automatically invalidated on first successful scan
- Non-transferable — if shared, later scan attempts are logged as duplicate/rejected
- Sent via email (image and optional PDF ticket); resend option for admins
- Bulk-upload generation and sending processed as a background batch job to handle thousands of records smoothly

5.4 Offline-Capable Scanner App (Gate Staff)

- Runs on gate staff's own phones as an installable web app — in-app scanner only, not the OS camera
- Downloads an encrypted, cached copy of the event's valid QR tokens ahead of time for fully offline validation
- Logs scan results locally (checked-in / duplicate / invalid) and queues them for sync
- On reconnect, syncs to the central server
- Conflict resolution: if the same code is scanned at multiple gates while offline, the earliest timestamp wins; later duplicate scans are automatically flagged as rejected once synced
- Each device session is tied to a specific gate staff login and, optionally, a named gate/entry point

5.5 Real-Time / Sync-Aware Admin Dashboard

- Live check-in feed as scans sync in, showing participant name and details per gate
- Dashboard clearly distinguishes synced vs. pending-sync counts, since some gates may be offline for stretches
- Aggregate stats: total registered, checked-in, remaining, and breakdown by gate

5.6 Messaging

- Email as the primary channel, with templated bulk/segmented messaging to registrants

- SMS/WhatsApp available as an optional add-on channel for reminders and time-sensitive updates
- Delivery logs maintained per campaign/message batch

5.7 Reporting & Exports

- CSV and PDF exports: full attendance list, no-shows, check-in-by-gate/time breakdown, and flagged/duplicate scan log
- Reports accessible per event and retained historically per client, supporting repeat and multi-event use

6. Functional Requirements

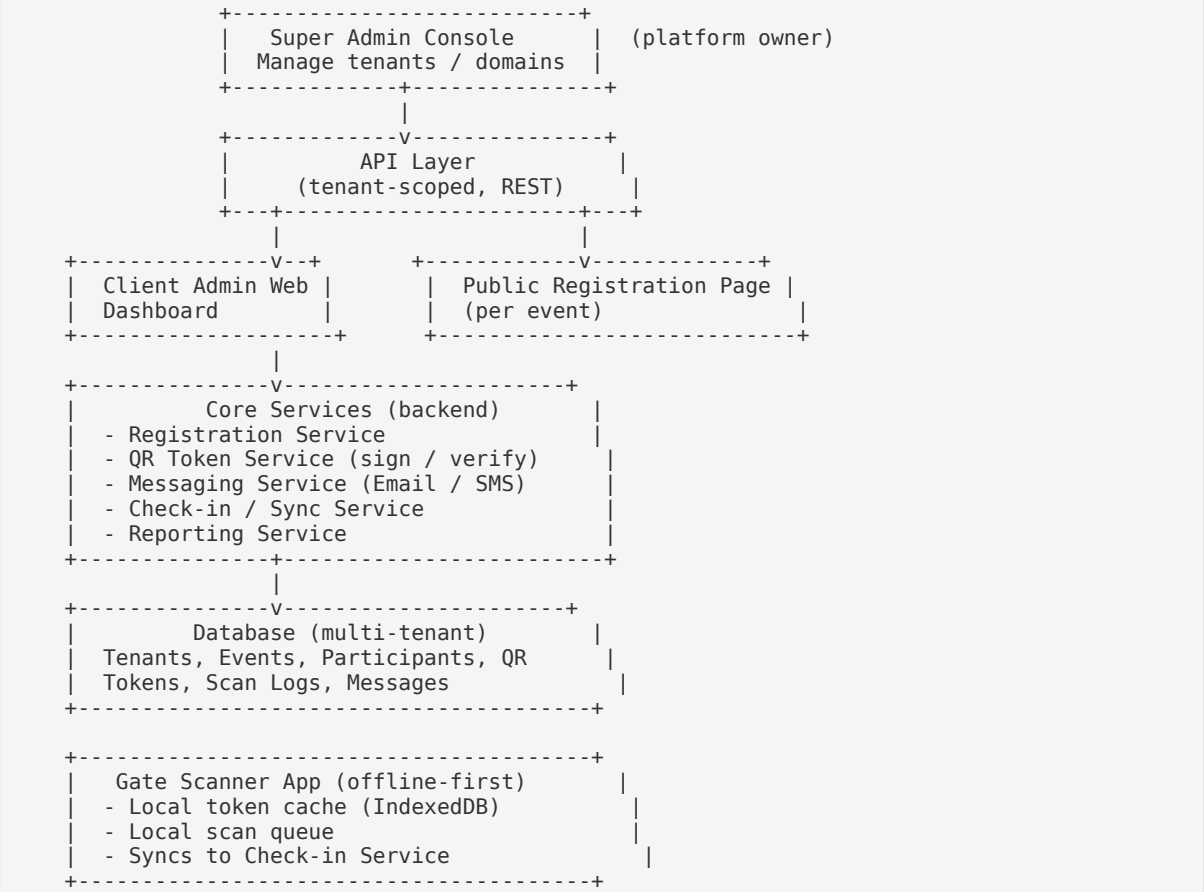
1. Platform shall support multiple isolated client tenants, each with a custom domain.
2. Each tenant shall support multiple independent events over time.
3. System shall generate a unique, non-sequential, signed QR code per registrant.
4. System shall support both self-registration and bulk Excel/CSV upload at scale (10,000+ rows).
5. System shall deliver QR codes via email automatically, with admin resend capability.
6. Scanner app shall validate QR codes fully offline using a pre-downloaded encrypted token set.
7. Scanner app shall support gate-scoped, scan-only logins distinct from full admin accounts.
8. System shall reconcile duplicate/conflicting offline scans using first-timestamp-wins on sync.
9. Dashboard shall reflect live and pending-sync check-in data with a clear distinction between the two.
10. System shall support segmented bulk email, with optional SMS/WhatsApp via a Nigeria-capable provider.
11. System shall generate CSV/PDF reports covering attendance, no-shows, and flagged scans.
12. System shall log every scan attempt with timestamp, gate/device ID, and result status.

7. Non-Functional Requirements

Category	Requirement
Security	Signed/encrypted QR tokens; strict tenant data isolation; NDPA-compliant handling of participant data and messaging consent
Performance	Must handle 10,000–15,000+ registrant records per event without slowdown; scan validation near-instant (~1 second), online or offline
Scalability	Multi-tenant, multi-event architecture designed in from v1, not retrofitted
Offline resilience	Core check-in function must work with zero connectivity at the venue for extended periods
Usability	Gate staff scanner interface usable with near-zero training, on personal phones of varying quality
Availability	High uptime during live event windows, particularly the first hour of gates opening (peak load)

8. System Architecture

The platform follows a multi-tenant service architecture with an offline-first gate scanner as a distinct client application. High-level component flow:



Key Architectural Decisions

- Tenant isolation is handled at the database layer (shared database with tenant_id scoping); this can move to schema-per-tenant later if a client requires stricter isolation.
- QR tokens are self-verifying (signed payload), so the scanner can validate authenticity offline without a server call. The server is needed only to check 'already used' status, which is why offline mode relies on a pre-synced token list.
- The sync service is the single source of truth for reconciling conflicts across gates, using first-timestamp-wins.

9. Data Model

Entity	Key Fields
Tenant	id, name, custom_domain, branding (logo/colors/sender email), plan, created_at
User	id, tenant_id, role (super_admin / client_admin / gate_staff), name, email, phone, password_hash, assigned_event_id, assigned_gate_name

Entity	Key Fields
Event	id, tenant_id, name, date, location, registration_form_schema, status (draft/live/closed), created_at
Participant	id, event_id, name, email, phone, custom_field_data, registration_source (self/bulk), registered_at, qr_token_id
QRToken	id, participant_id, signed_payload, status (unused/used/revoked), used_at, used_by_gate_id, created_at
ScanLog	id, qr_token_id, event_id, gate_staff_id, gate_name, device_id, scan_result (success/duplicate/invalid), scanned_at, synced_at, is_conflict_resolved_loser
MessageCampaign	id, event_id, channel (email/sms/whatsapp), segment, template, sent_count, status, created_at
MessageLog	id, campaign_id, participant_id, channel, delivery_status, sent_at
Report/Export	id, event_id, type (attendance/no-show/gate-breakdown/flagged), format (csv/pdf), generated_at, file_url

10. Key API Endpoints (Representative)

```
POST   /events/:id/register              -> self-registration
POST   /events/:id/participants/bulk     -> bulk upload (CSV/Excel)
GET    /events/:id/qr-tokens/export      -> encrypted token bundle for scanner pre-download
POST   /scan/validate                    -> online validation (fallback when connected)
POST   /scan/sync                        -> batch upload of offline scan logs
GET    /events/:id/dashboard             -> live/pending check-in stats
POST   /events/:id/messages              -> send bulk/segmented message
GET    /events/:id/reports/:type         -> generate/download report
POST   /tenants                          -> (super admin) create new client tenant
```

11. Key Screens

11.1 Public Registration Page (tenant-branded, per event)

- Header: client logo plus event name, date, and location
- Form: name, email, phone, and custom fields defined by the admin
- Submission leads to a confirmation screen: "Check your email for your entry QR code"

11.2 Client Admin Dashboard — Overview

- Top stats bar: total registered, checked-in, remaining, percent checked-in
- Live feed panel: scrolling list of most recent check-ins (name, gate, time)
- Gate breakdown mini-chart: check-ins per gate
- Pending-sync indicator if any gate devices have not synced recently

11.3 Client Admin — Participants Table

- Columns: name, email, phone, registered (with timestamp), checked-in (with timestamp and gate), source
- Actions: resend QR, export, bulk upload, search/filter

11.4 Client Admin — Messaging

- Compose panel: choose channel (email/SMS/WhatsApp), segment, and template
- Preview and send
- History table of past campaigns with delivery stats

11.5 Gate Scanner App (Mobile, Gate Staff)

- Full-screen in-app camera scan view (not the device's native camera)
- Valid scan: green confirmation with participant name and "Checked In"
- Duplicate scan: amber warning showing where and when it was first used
- Invalid scan: red error for an unrecognized code
- Status bar showing offline queue count or sync confirmation
- Gate name displayed at the top, tied to the logged-in device

11.6 Reports Screen

- List of available reports per event: attendance, no-shows, gate breakdown, flagged/duplicates
- Format toggle between CSV and PDF
- Download action per report

11.7 Super Admin Console

- List of client tenants with their custom domains, plan, and usage (events run, participants processed)
- "Add New Client" flow: name, domain setup, initial admin invite

12. Suggested Technology Approach

Layer	Recommendation
Backend	Multi-tenant architecture (tenant-scoped data), REST/GraphQL API
Client Admin Web App	Responsive dashboard covering event setup, bulk upload, messaging, live monitoring, and reports
Gate Scanner App	Installable web app (PWA) on gate staff's own phones, with local storage (IndexedDB) for offline token cache and scan queue
QR Tokens	Signed payloads (e.g., HMAC-signed JSON encoded in the QR) so validity can be verified offline without a server call
Email	Transactional email provider (e.g., SendGrid, Mailgun, or Amazon SES) with batch sending for bulk uploads
SMS/WhatsApp	Termii recommended for Nigeria; Twilio as a fallback if expanding beyond Nigeria
Custom Domains	Wildcard/subdomain routing per tenant, with optional fully custom domain via CNAME
Reporting	Server-side report generation, exportable as CSV and PDF

13. Phased Rollout Plan

Phase 1 — MVP

- Single-tenant core: registration (self-service and bulk)
- QR generation and email delivery
- Offline scanner with sync
- Basic dashboard
- CSV export

Phase 2 — White-Label Layer

- Multi-tenant/white-label architecture
- Custom domains per client
- Gate-staff role separation
- PDF reports

Phase 3 — Messaging & Scale

- SMS/WhatsApp integration (Termii)
- Advanced reporting and analytics
- Super Admin console for managing multiple clients

This phased approach validates the core check-in experience at scale before layering on the full white-label business model.

14. Approval

This document reflects the requirements and design approved for the Event Check-in Platform as of the date on the cover page. Any material changes to scope, architecture, or rollout phasing should be reflected in a revised version of this document.